

Income Classification with Naïve Bayes

Report

Introduction

This project focuses on implementing a Mixed Naïve Bayes classifier to predict an individual's annual income as either high income (over \$50000) or low income (\$50000 or less). The project applies and compares both supervised and semi-supervised learning techniques. Three datasets were used which include, the supervised training set, unlabelled training set and the test set. All data is derived from the 1994 US Census.

Question 1

Table 1: Class Prior Probability

Class	Prior Probability
<=50K	0.754
>50K	0.246

The prior probabilities show class imbalance since, 75.4% of instances belong to the low income class and only 24.6% belong in the higher income class. This creates bias in the Naïve Bayes classifier towards the majority class. This results in better performance for the majority but poor recall for the minority class.

Table 2: Continuous Feature Means

Feature	Low Income Mean	High Income Mean
age	37.05	43.94
education-num	9.63	11.59
capital-gain	157.67	3607.15
capital-loss	55.97	202.36
hours-per-week	39.43	45.64

The mean value table shows the largest differences in capital-gain and capital-loss. This is expected, as higher income earners are more likely to participate in investment opportunities.

Table 3: Continuous Feature Standard Deviations (SD)

Feature	Low Income SD	High Income SD
age	13.71	10.3
education-num	2.45	2.36
capital-gain	1017.86	13616.62
capital-loss	316.02	603.99
hours-per-week	11.91	10.4

However, the standard deviations indicate high variability within each class, making both these features less likely to consistently distinguish between the classes. Therefore, Cohen's d is used for a fair comparison across all features with different scales and variances (Daines, 2024).

Table 4: Separation

Feature	Cohen's d
education-num	0.81
hours-per-week	0.54
age	0.53
capital-gain	0.51
capital-loss	0.36

Education-num has the highest d-value and is therefore the strongest separator, followed by hours-per-week. These observations are intuitive as higher education levels generally create pathways to higher paying roles, and most organisations pay on an hourly basis.

Table 5: Most Predictive Category Value Per Feature

Feature	Class	Category	R
workclass	>50K	Self-emp-inc	3.25
workclass	<=50K	Private	0.86
education	>50K	Prof-school	7.96
education	<=50K	9th	0.15
marital-status	>50K	Married-AF-spouse	4.29
marital-status	<=50K	Never-married	0.16
occupation	>50K	Exec-managerial	2.77
occupation	<=50K	Priv-house-serv	0.08
relationship	>50K	Wife	2.74
relationship	<=50K	Own-child	0.05
race	>50K	Asian-Pac-Islander	1.13
race	<=50K	Amer-Indian-Eskimo	0.44
sex	>50K	Male	1.38
sex	<=50K	Female	0.4
native-country	>50K	Taiwan	3.38
native-country	<=50K	Mexico	0.18

Table 6: 5 Most Predictive Category Values for High Income Class

Feature	Category	R
education	Prof-school	7.96
education	Doctorate	7.14
marital-status	Married-AF-spouse	4.29
education	Masters	3.56
native-country	Taiwan	3.38

Table 7: 5 Most Predictive Category Values for Low Income Class

Feature	Category	R
relationship	Own-child	0.05
occupation	Priv-house-serv	0.08
occupation	Other-service	0.14
relationship	Other-relative	0.14
education	9th	0.15

The above results display education, occupation and relationship are the strongest predictive categorical features. Advanced qualifications such as Prof-school and Doctorate are strongly associated with higher income, while lower education levels such as 9th grade are linked to low income. Occupation associates high-income individuals with professional roles such as Exec-managerial, and low-income individuals to lower-skill jobs like Priv-house-serv. Economically dependent individuals such as Own-child are associated with low income whereas, relationships

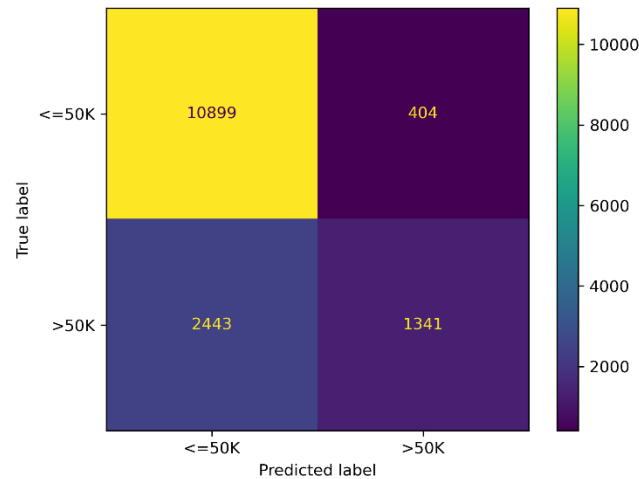
such as Wife are associated with higher income. The learned parameters appear to be reasonable. Education and education_num are the features expected to most useful for classification as they lead to the best separation and have the strongest predictive power across both classes.

Question 2

Table 8: Supervised Model Classification Report

	precision	recall	f1-score	support
<=50K	0.817	0.964	0.884	11303
>50K	0.768	0.354	0.485	3784
accuracy	0.811	0.811	0.811	0.811
macro avg	0.793	0.659	0.685	15087
weighted avg	0.805	0.811	0.784	15087

Figure 1: Supervised Model Confusion Matrix



The model has a relatively high accuracy and can correctly predict the class 81.1% of the time. However, as expected, due to class imbalance, all the performance metrics drop for the minority class, specifically recall. The confusion matrix highlights the significant misclassification, with 2443 high-income individuals incorrectly being predicted as low income.

Table 9: Test instances with at least one unseen category value

Total
Unseen Category Values1

Unseen category values are handled through two approaches. Firstly, for category values that were present during training but not seen in any instances, Laplace smoothing is applied (GeeksforGeeks, 2024). This prevents zero probabilities and improves the generalisability of the model. Secondly, the category values not present during training are encoded as the first category value for that feature. This ensures the predict_log_proba function works without errors and handles this rare case of inconsistency with simplicity.

Table 10: Number of Instances Skipped

	Total
Test Instances	15087
Prediction Instances	15087

All instances were successfully classified, as rows with missing values were removed during preprocessing and unseen category values were handled appropriately.

Table 11: Top 3 Confident Instances for High Income Class

age	workclass	education	education-num	marital-status	occupation	relationship	race	sex	capital-gain	capital-loss	hours-per-week	native-country	R
50	Self-emp-not-inc	Prof-school	15	Married-civ-spouse	Exec-managerial	Husband	White	Male	99999	0		50 United-States	2.69E+43
42	Self-emp-not-inc	Masters	14	Married-civ-spouse	Sales	Husband	White	Male	99999	0		80 United-States	2.69E+43
52	Private	Assoc-voc	11	Married-civ-spouse	Prof-specialty	Husband	White	Male	15024	0		40 United-States	2.69E+43

The high income predictions show high capital gains, longer working hours and higher education standards, which are true for wealthier demographics.

Table 12: Top 3 Confident Instances for Low Income Class

age	workclass	education	education-num	marital-status	occupation	relationship	race	sex	capital-gain	capital-loss	hours-per-week	native-country	R
18	Private	10th	6	Never-married	Other-service	Own-child	White	Female	0	0		3 United-States	9.83E-11
17	Private	10th	6	Never-married	Other-service	Own-child	Black	Female	0	0		12 United-States	1.14E-10
17	Private	11th	7	Never-married	Other-service	Own-child	White	Female	0	0		5 United-States	1.18E-10

Low income predictions show younger individuals with fewer working hours and lower education levels, which is consistent with students who often work part-time.

Table 13: 3 Near Decision Boundary Instances

age	workclass	education	education-num	marital-status	occupation	relationship	race	sex	capital-gain	capital-loss	hours-per-week	native-country	R
38	Private	Bachelors	13	Never-married	Exec-managerial	Not-in-family	White	Male	0	1340		40 United-States	0.997772363
37	Self-emp-not-inc	Prof-school	15	Married-civ-spouse	Prof-specialty	Husband	White	Male	0	0		40 Cuba	0.99724831
61	Private	Masters	14	Married-civ-spouse	Exec-managerial	Husband	White	Male	0	0		60 United-States	0.996298459

Low-confidence predictions are difficult to categorise due to conflicting attributes, for example, high level education and occupations paired with zero capital gains.

Table 14: Mean Posterior Ratios

Class	Mean_R
<=50K	0.032
>50K	4.58E+42

The mean posterior ratio for the higher income class is extremely large. This means that the model must be highly confident to be able to predict the minority class. In contrast, the low income class has a mean closer to 1, showing that the model is more lenient for the majority class, making it easier to predict.

Question 3

For this project, label propagation (Option 1) was selected as the semi-supervised learning method. Before extending the supervised model through this technique, its dataset was split into 80% for training and 20% for validation. This split maximised the data available for training, while also ensuring the validation set was large enough to accurately evaluate the model's performance. Stratification was applied to ensure the training and validation datasets maintain the same class balance as the original dataset.

Table 15: Model A Accuracy

Model	Validation Accuracy
A	0.786

Model A was trained on the combined supervised and pseudo-labelled data to utilise the unlabelled data and the observed accuracy for Model A on the validation. As this was the simplest extension to the supervised model, no hyperparameter tuning was performed.

Table 16: Model B Hyperparameter Tuning

Confidence Percentile	Unlabelled Data Kept (%)	Validation Accuracy
10	89.9993	0.7875
15	84.999	0.7885
20	79.9987	0.7888
25	74.9983	0.7888
30	69.998	0.7895
35	64.9977	0.7921

Model B was trained using both supervised and high-confidence pseudo-labelled data. To filter the confident predictions, percentiles were used to remove a certain percentage of the least confident predictions. The optimal percentage of low confidence predictions to remove, which led to the best accuracy, was determined within the range of 10-35% through hyperparameter tuning. This range was specifically chosen to avoid intense filtering, as discarding a greater amount of unlabelled data would diminish the value of semi-supervised learning. The best performing Model B retained 65% of the pseudo-labelled data and achieved an accuracy of 79.2% on the validation set. Thus, outperforming Model A. This suggests that Model B successfully reduced the impact of unreliable predictions, improving the quality of training data and outweighing the loss of training instances.

Table 17: Model C Hyperparameter Tuning

Unlabelled Split (%)	Validation Accuracy
30	0.7855
40	0.7855
50	0.7855
60	0.7855

Model C performs iterative pseudo-labelling in two stages and retrain the model after each stage. During hyperparameter tuning, different split ratios (30-70, 40-60, 50-50, 60-40) were tested. This moderate range is chosen to ensure the model does not overfit due to a small subset of data in the first iteration while still leaving enough data to expand its understanding in the second iteration. To prevent ordering bias, the unlabelled training set is reshuffled once to randomly distribute the instances before applying the split. The validation set performance showed consistent accuracy of 78.5% across the different hyperparameter values, which is lower than both Model A and Model B. This suggests the model is insensitive to how pseudo-labelled data is partitioned. Additionally, the iterative label propagation doesn't improve the model and actually decreases its accuracy by reinforcing and propagating errors across iterations.

The overall performance of the three models was similar, with only minor differences in the validation accuracy. However, Model B was selected as the best semi-supervised model since it achieved the highest accuracy. This final model was then retrained using the full supervised training set dataset and the selected pseudo-labelled instances to ensure it could benefit from all available training data (Jainvidip, 2024). This ensured the final model could utilise all available training data, further increasing its ability to detect the underlying patterns.

Question 4

Table 18: Model Performance Metrics Comparison

Model	Accuracy	Precision	Recall	F1
Supervised Model	0.8113	0.7685	0.3544	0.4851
Semi-Supervised Model	0.7981	0.7055	0.3348	0.4541

After evaluating the final semi-supervised model on the test data, it was found that all its performance metrics were lower than the supervised model. The biggest difference was in their accuracy score, which differs by 1.3%. This result is expected because treating pseudo-labels as ground truth can introduce confirmation bias (Vanhoucke, 2019). By training on its own incorrect but high-confidence predictions, the model produces a higher number of false negatives and false positives. In contrast, the supervised model learns from the true labels and therefore has better overall performance.

Table 19: Posterior Ratio Distribution Comparison

Model	Mean	Median	Standard Deviation	IQR	Skewness
Supervised Model	5.29623E+41	0.0004	3.7319E+42	0.01874	6.91234
Semi-Supervised Model	6.14708E+41	0.00006	4.01821E+42	0.00476	6.38387

The distribution of the posterior ratios has shifted significantly across the two models. Previously, the posterior ratios for the high income class were right-skewed, but this skewness has now reduced. Therefore, the distribution has been pulled closer to the decision boundary. This means that the model no longer needs to be as certain as before to predict the minority class. However, more instances are now being classified near the decision boundary, increasing the model's uncertainty. The smaller IQR further supports these results. The median has gotten closer to 1 and the standard deviation has increased. This further highlights the ambiguity being faced by the model as well as the increased variability in its decisions. Finally, the mean of the semi-supervised model is higher, possibly due to some extreme values influencing the distribution.

Table 20: Change in Means

Feature	Class	Mean Shift	Inc/Dec
age	<=50K	-0.7731	Decrease
age	>50K	0.2413	Increase
education-num	<=50K	-0.1237	Decrease
education-num	>50K	0.1094	Increase
capital-gain	<=50K	-55.3556	Decrease
capital-gain	>50K	2457.3063	Increase
capital-loss	<=50K	-23.5951	Decrease
capital-loss	>50K	124.7155	Increase
hours-per-week	<=50K	-0.5672	Decrease
hours-per-week	>50K	0.1599	Increase

The above results show meaningful changes for some feature parameters between the two models. For continuous features, capital gain and capital loss significantly increased for the high income class, while they decreased for the low income class. This suggests that the semi-supervised model has become more reliant on these features to distinguish between the two classes.

Table 21: Change in Variances

Feature	Class	Variance Change	Inc/Dec
age	<=50K	-0.2168	Decrease
age	>50K	4.4213	Increase
education-num	<=50K	-0.0944	Decrease
education-num	>50K	0.04475	Increase
capital-gain	<=50K	-424146.9135	Decrease
capital-gain	>50K	129934613.3	Increase
capital-loss	<=50K	-41967.4748	Decrease
capital-loss	>50K	193591.9372	Increase
hours-per-week	<=50K	0.3029	Increase
hours-per-week	>50K	5.6864	Increase

Additionally, the variance of hours-per-week has increased for the high income class and decreased for the low income class. This suggests that high-income individuals have a wider range of working hours, while the low income class is more concentrated around a specific range of working hours.

Table 22: Change in Category Probabilities

Feature	Class	LogProb Change	Direction
workclass	<=50K	-0.0953	Decrease
workclass	>50K	-0.0326	Decrease
education	<=50K	-0.0293	Decrease
education	>50K	-0.0395	Decrease
marital-sta	<=50K	0.0629	Increase
marital-sta	>50K	-0.0104	Decrease
occupation	<=50K	0.0017	Increase
occupation	>50K	-0.0038	Decrease
relationshi	<=50K	0.0322	Increase
relationshi	>50K	0.0273	Increase
race	<=50K	0.0444	Increase
race	>50K	0.0048	Increase
sex	<=50K	0.0127	Increase
sex	>50K	0.0008	Increase
native-cou	<=50K	-0.0601	Decrease
native-cou	>50K	-0.0225	Decrease

For the categorical features, the log-probabilities increased for marital-status and race in the low income class, increasing their role in identifying low-income instances in the semi-supervised model. The relationship features also saw an increase in log-probabilities for both income classes, suggesting the semi-supervised model is more sensitive to personal dynamics.

Table 23: Supervised Model Top 5 Predictors for High Income Class

Feature	Category	R
education	Prof-school	7.96
education	Doctorate	7.14
marital-status	Married-AF-spouse	4.29
education	Masters	3.56
native-country	Taiwan	3.38

Table 24: Semi-Supervised Model Top 5 Predictors for High Income Class

Feature	Category	R
education	Prof-school	13.41
education	Doctorate	11.72
workclass	Self-emp-inc	4.8
education	Masters	4.52
native-country	Hungary	4.2

Table 25: Supervised Model Top 5 Predictors for Low Income Class

Feature	Category	R
education	9th	0.15
relationship	Other-relative	0.14
occupation	Other-service	0.14
occupation	Priv-house-serv	0.08
relationship	Own-child	0.05

Table 26: Semi-Supervised Model Top 5 Predictors for Low Income Class

Feature	Category	R
education	9th	0.13
relationship	Other-relative	0.12
occupation	Other-service	0.12
occupation	Priv-house-serv	0.08
relationship	Own-child	0.05

Minor differences were observed in the most predictive features. For the low income class, the best categorical predictors remained unchanged, while workclass replaced marital-status in the hierarchy for high income predictors.

Table 27: Supervised Model Separation

Feature	Cohen's d
education-num	0.81
hours-per-week	0.54
age	0.53
capital-gain	0.51
capital-loss	0.36

Table 28: Semi-Supervised Model Separation

Feature	Cohen's d
education-num	0.91
capital-gain	0.76
capital-loss	0.75
age	0.6
hours-per-week	0.59

The top continuous predictors remain education-num, age and hours-per-week, but in a different order.

Conclusion

In essence, despite the observed class imbalance in the training dataset, the supervised Naïve Bayes model demonstrated strong overall performance. Among the three semi-supervised models, Model B performed the best. However, it still could not outperform the supervised model. This suggests that, in this case, the supervised model was more effective than its extension that used label propagation. Other supervised learning techniques, such as active learning and expectation-maximisation, could also be explored to further improve the model.

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